

Data Analytics Internship

Project 1: Data Cleaning & Preparation

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Goal

Clean a raw e-commerce sales dataset by handling missing values, duplicates, and incorrect data formats, transforming it into a reliable, analysis-ready dataset.

Key Requirements

- Identify missing or null values
- Remove duplicates
- Correct data formats (dates, numbers, text)

Dataset Overview

The dataset contained 1,200 order records across 14 columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice.

What Was Done

1. Removed all duplicate OrderIDs, confirming 100% unique records across the dataset.
2. Standardized all dates to a consistent YYYY-MM-DD format across the Date column.
3. Filled blank CouponCode cells with "NONE" to preserve record completeness without losing context.
4. Cleaned up whitespace issues, removing leading and trailing spaces across text columns.
5. Verified every change using formula-based quality checks before finalizing the dataset.

Verification Results

Check	Formula	Result
Missing CouponCode values	=COUNTBLANK(L2:L1201)	0
Whitespace in Date column	=SUMPRODUCT(--(D2:D1201<>TRIM(D2:D1201)))	0

Both verification checks returned 0, confirming zero missing values in the CouponCode column and zero whitespace inconsistencies in the Date column.

Conclusion

This project reinforced the foundation of good data analysis: clean, reliable data is everything. By systematically identifying and resolving missing values, duplicates, and formatting issues, the dataset was transformed from raw, inconsistent records into a verified, analysis-ready resource. This clean dataset became the foundation for the Exploratory Data Analysis carried out in Project 2.

Tools Used

Excel, Google Sheets, formula-based data verification.